

Random variables and probability distributions

Covered by chapter 2.1-2.4 in the textbook.

Definition: Random variable is a variable which get its numerical value determined by the outcome of a stochastic experiment. $\square$

Examples: Amount of oil found in a core sample, lifetime of an electronic component, the temperature tomorrow at 12 a.m., the number of bugs in a new software, the number of accidents on a road, the result of a throw of a dice, ... (most variables can be considered as random variables). $\square$

Random variables are either discrete (e.g. number of bugs, number of accidents, throw of dice) or continuous (e.g. oil content, life time, temperature).

A discrete random variable can only attain a countable set of values, while a continuous variable can attain any value in an interval.

Notation: Random variables are denoted by capital letters, X , Y , Z , etc (“before the experiment”), while the numerical value of the random variable (“after the experiment”) are denoted by lower case letters, x , y , z , etc. $\square$

Discrete and continuous r.v.s (random variables) must be treated differently. For discrete values it is meaningful to talk about $P(X = a)$, while for continuous variables $P(X = a) = 0$ always. For continuous variables we only consider the probability of getting values in an interval, e.g. $P(a \leq X \leq b)$.

Discrete probability distributions

Definition: $f(x)$ is the probability mass function (pmf) of the discrete r.v. X if for all possible outcomes x :

1. $f(x) \geq 0$
2. $\sum_x f(x) = 1$
3. $f(x) = P(X = x)$ □

The book also use the notation $p_X(x)$ for pmf, i.e. $p_X(x) = f(x)$.

Example: Throw a coin three times. Let X ="number of heads".

Sample space	x	Sample space	x
HHH	3	HTT	1
HHT	2	THT	1
HTH	2	TTH	1
THH	2	TTT	0

We see that:

$$f(0) = P(X = 0) = 1/8$$

$$f(1) = P(X = 1) = 3/8$$

$$f(2) = P(X = 2) = 3/8$$

$$f(3) = P(X = 3) = 1/8$$

This is the pmf (probability mass function) of X . □

See R-code: `number_of_heads.R`

Definition: The cumulative distribution function (cdf) of a discrete r.v. X is:

$$F(x) = P(X \leq x) = \sum_{t \leq x} f(t) \quad \square$$

Continuous probability distributions

Definition: $f(x)$ is the probability density function (pdf) of the continuous r.v. X if:

1. $f(x) \geq 0$ for all $x \in \mathcal{R}$
2. $\int_{-\infty}^{\infty} f(x)dx = 1$
3. $P(a < X < b) = \int_a^b f(x)dx$ □

Note:

$$\text{Discrete: } P(a \leq X \leq b) = \sum_{a \leq x \leq b} f(x)$$

$$\text{Continuous: } P(a \leq X \leq b) = \int_a^b f(x)dx$$

Also note that for discrete r.v.s:

$$P(X \leq a) \neq P(X < a) \text{ when } P(X = a) > 0,$$

while for continuous r.v.s:

$$\begin{aligned} P(X \leq a) &= P(X < a) + P(X = a) \\ &= P(X < a). \end{aligned}$$

Example: Let Y ="lifetime of an electronic component". Assume that it is known that Y has the pdf (probability density function):

$$f(y) = \begin{cases} \frac{1}{5}e^{-y/5} & ,y \geq 0 \\ 0 & ,\text{otherwise} \end{cases}$$

(Check by yourself that $\int_{-\infty}^{\infty} f(y)dy = 1$).

We can now calculate probabilities, e.g.:

$$\begin{aligned} P(Y \leq 5) &= \int_{-\infty}^5 f(y)dy = \int_0^5 \frac{1}{5}e^{-y/5}dy \\ &= -e^{-1} + e^0 = \underline{\underline{0.63}} \end{aligned}$$

$$\begin{aligned} P(4 < Y < 5) &= \int_4^5 f(y)dy = \int_4^5 \frac{1}{5}e^{-y/5}dy \\ &= -e^{-1} + e^{-4/5} = \underline{\underline{0.08}} \quad \square \end{aligned}$$

See R-code: `exponential_distribution_demo.R`

Definition: The cumulative distribution function (cdf) of a continuous r.v. X is:

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t)dt \quad \square$$

The lifetime example:

$$F(y) = \int_{-\infty}^y f(t)dt = \int_0^y \frac{1}{5} e^{-t/5} dt = 1 - e^{-y/5}$$

$$F(5) = 1 - e^{-5/5} = \underline{\underline{0.63}} \quad \square$$

Generally: The distribution of a random variable is specified by the pmf/pdf (or the cdf).

Joint distribution

Joint distribution = the distribution of two or more r.v.s considered simultaneously.

Definition: If X and Y are two *discrete* r.v.s then $f(x, y)$ is the joint pmf of X and Y if:

1. $f(x, y) \geq 0$
2. $\sum_x \sum_y f(x, y) = 1$
3. $f(x, y) = P(X = x, Y = y)$ $\square$

Definition: If X and Y are two *continuous* r.v.s then $f(x, y)$ is the joint pdf of X and Y if:

1. $f(x, y) \geq 0$
2. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$
3. $P((X, Y) \in A) = \int \int_A f(x, y) dx dy$ $\square$

Example: Plant/soil example Let

X = relative water consumption of a plant

Y = relative humus content of the soil

Assume that X and Y has joint pdf

$$f(x, y) = \begin{cases} x - \frac{1}{2}y + \frac{3}{4} & , 0 \leq x \leq 1, \quad 0 \leq y \leq 1 \\ 0 & , \text{otherwise} \end{cases}$$

(Check yourself: $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$).

Then for instance:

$$\begin{aligned} & P(X < 0.5, Y > 0.75) \\ &= \int_{0.75}^1 \int_0^{0.5} \left(x - \frac{1}{2}y + \frac{3}{4}\right) dx dy = \underline{\underline{\frac{9}{128}}} \quad \square \end{aligned}$$

Definition: Marginal distributions (=ordinary probability distributions) for X or Y alone are found from the joint distributions by summing/integrating out the other variable. For instance are the marginal distributions of X given by $g(x) = \sum_y f(x, y)$ if X and Y are discrete and by $g(x) = \int_{-\infty}^{\infty} f(x, y) dy$ if X and Y are continuous . $\square$

Example: Plant/soil example. X =relative water consumption of a plant

$$\begin{aligned} g(x) &= \int_{-\infty}^{\infty} f(x, y) dy = \int_0^1 (x - \frac{1}{2}y + \frac{3}{4}) dy \\ &= x + \frac{1}{2}, 0 \leq x \leq 1 \quad \square \end{aligned}$$

Conditional distributions

Remember:

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \quad , \text{if } P(A) > 0$$

Then it follows that if X and Y are discrete r.v.s:

$$\begin{aligned} P(Y = y|X = x) &= \frac{P(X = x, Y = y)}{P(X = x)} \\ &= \frac{f(x, y)}{g(x)} \quad , g(x) > 0 \end{aligned}$$

A corresponding relation holds for continuous r.v.s.

Definition: If X and Y are two r.v.s (discrete or continuous) the conditional pmf/pdf of Y given $X = x$ (i.e. when we *know* the value of X):

$$f(y|x) = \frac{f(x, y)}{g(x)} \quad , g(x) > 0$$

Similarly the conditional distribution of X given $Y = y$ is:

$$f(x|y) = \frac{f(x, y)}{h(y)}, h(y) > 0 \quad \square$$

Example: Water. Assume that we know that the water consumption is 0.5. What is the probability that the humus content is less than 0.4? We have that

$$f(y|x) = \frac{f(x, y)}{g(x)} = \frac{x - 0.5y + 0.75}{x + 0.5}$$

Then:

$$\begin{aligned} P(Y < 0.4|X = 0.5) \\ &= \int_0^{0.4} f(y|0.5)dy \\ &= \int_0^{0.4} \frac{0.5 - 0.5y + 0.75}{0.5 + 0.5}dy \\ &= \int_0^{0.4} (1.25 - 0.5y)dy = \underline{\underline{0.46}} \end{aligned}$$

(Is for instance $P(Y < 0.4|X = 0.75)$ larger or smaller? Calculate yourself!). $\square$

If $f(x|y)$ does not depend on y (or correspond-

ing $f(y|x)$ does not depend on x) then X and Y are independent.

Definition: Two r.v.s X and Y are independent if and only if

$$f(x, y) = g(x) \cdot h(y) \quad , \text{for all } x, y \quad \square$$

In practice independence means that information about the value of one of the variables does not have any influence on the distribution of the other variable.

Example: In the water example we see that X and Y are not independent (they are in other words *dependent*) since $f(y|x)$ depends on x (is a function of x).

Expectation and variance

Expectation

Expectation = “the long run average”.

Definition: The expectation of a random variable X is:

$$E(X) = \begin{cases} \sum_x x f(x) & , X \text{ discrete} \\ \int_{-\infty}^{\infty} x f(x) dx & , X \text{ continuous} \end{cases}$$

We often use μ or μ_X as symbol for $E(X)$. $\square$

Example: Throw a coin 3 times. Let X = “number of heads”. We have shown earlier that:

x	0	1	2	3
$f(x)$	1/8	3/8	3/8	1/8

We get that

$$E(X) = \sum_x x f(x) = 0 \cdot \frac{1}{8} + 1 \cdot \frac{3}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{1}{8} = \underline{\underline{\frac{3}{2}}} \quad \square$$

Example: Y = "lifetime of an electronic component". Y has pdf

$$f(y) = \begin{cases} \frac{1}{5}e^{-y/5} & , y \geq 0 \\ 0 & , \text{otherwise} \end{cases}$$

which gives that

$$\begin{aligned} E(Y) &= \int_{-\infty}^{\infty} yf(y)dy = \int_0^{\infty} y\frac{1}{5}e^{-y/5}dy \\ &= [-ye^{-y/5}]_0^{\infty} - \int_0^{\infty} -e^{-y/5}dy \\ &= 0 - [5e^{-y/5}]_0^{\infty} = \underline{\underline{5}} \end{aligned}$$

I.e., if we use many such components we will find that they on average have a lifetime of around 5. □

The expectation of functions of random variables are found similarly:

Definition:

$$E(g(X)) = \begin{cases} \sum_x g(x)f(x) & , X \text{ discrete} \\ \int_{-\infty}^{\infty} g(x)f(x)dx & , X \text{ continuous} \end{cases}$$

Example: The definition of variance on the next page is an important special case.

Definition: If X and Y are two r.v.s with joint pmf/pdf $f(x, y)$, then

$$E(g(X, Y)) = \begin{cases} \sum_x \sum_y g(x, y)f(x, y) & , \text{ disc.} \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y)f(x, y)dxdy & , \text{ cont.} \end{cases}$$

The definition of covariance on page 19 is an important special case. The rules for linear combinations given on page 25 also originate from this basic definition.

Variance and covariance

Variance = a measure of variation around the expectation.

Definition: The variance of a r.v. X is:

$$\begin{aligned}\sigma^2 &= \sigma_X^2 = \text{Var}(X) = \text{E}[(X - \mu)^2] \\ &= \begin{cases} \sum_x (x - \mu)^2 f(x) & , X \text{ discrete} \\ \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx & , X \text{ continuous} \end{cases}\end{aligned}$$

$\sigma = \text{SD}(X) = \sqrt{\text{Var}(X)}$ is called the standard deviation of X . □

Equivalent:

$$\text{Var}(X) = \text{E}(X^2) - \mu^2 = \text{E}(X^2) - \text{E}(X)^2 \quad \square$$

Example: Y = "lifetime of an electronic component".

$$\begin{aligned}\text{E}(Y^2) &= \int_0^{\infty} y^2 \frac{1}{5} e^{-y/5} dy \\ &= \dots = 50\end{aligned}$$

$$\Rightarrow \text{Var}(Y) = \text{E}(Y^2) - \text{E}(Y)^2 = 50 - 5^2 = \underline{\underline{25}} \quad \square$$

Rule of thumb: Often (but not always...) around 95% of the observations will fall in the interval $[\mu - 2\sigma, \mu + 2\sigma]$.

Definition: The covariance of X and Y is given by:

$$\begin{aligned}\sigma_{XY} &= \text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] \\ &= E(XY) - E(X)E(Y) \\ &= \begin{cases} \sum_x \sum_y xyf(x, y) - \mu_X \mu_Y & , \text{disc.} \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xyf(x, y) dx dy - \mu_X \mu_Y & , \text{cont.} \end{cases}\end{aligned}$$

Covariance = a measure of relationship,
 $\text{Cov}(X, Y) > 0$, “positive” relationship between X and Y (e.g. height/shoe size), $\text{Cov}(X, Y) < 0$, “negative” relationship between X and Y (e.g. age of care/value of car).

Note: X and Y independent $\Rightarrow \text{Cov}(X, Y) = 0$.

Also note: The opposite is not the case. Two variables can have $\text{Cov}(X, Y) = 0$ and still be dependent!

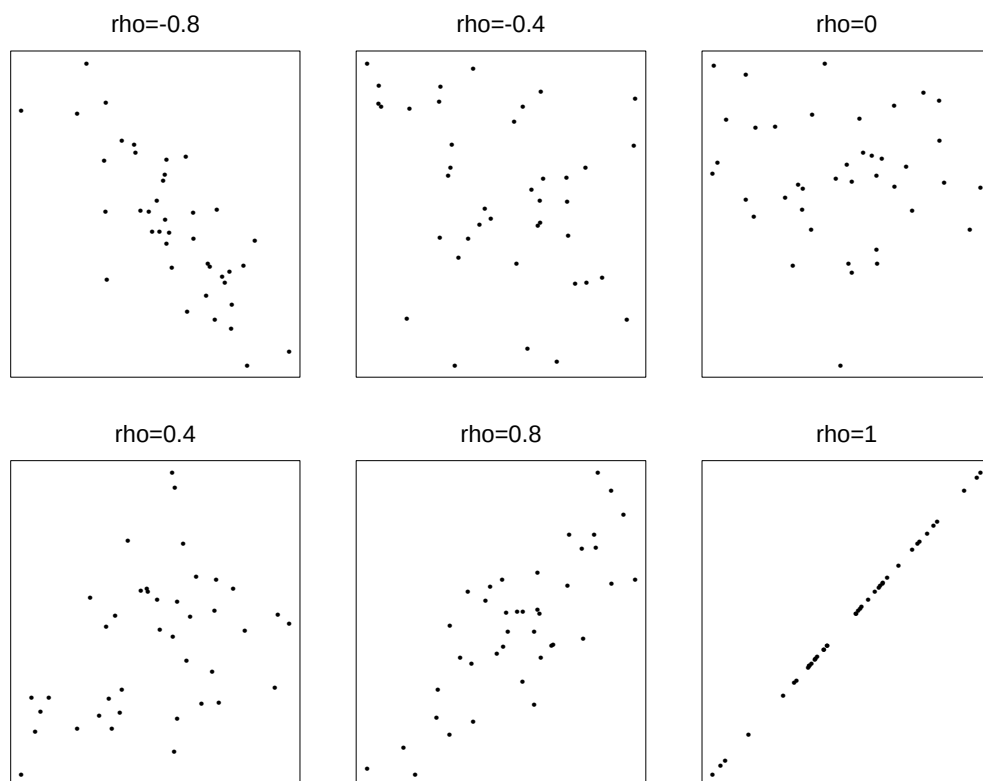
Problem: The covariance is often difficult to interpret, depends on the scale used. Instead we often use the correlation which is a normalization of the covariance.

Definition: Correlation

$$\rho(X, Y) = \text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

$$-1 \leq \rho(X, Y) \leq 1$$

Correlation = degree of linear relationship between X and Y :



Example: Water, X = “relative water consumption”, Y = “relative humus content”.

$$f(x, y) = \begin{cases} x - \frac{1}{2}y + \frac{3}{4} & , 0 \leq x \leq 1, \quad 0 \leq y \leq 1 \\ 0 & , \text{otherwise} \end{cases}$$

Find $\text{Corr}(X, Y)$:

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f(x, y) dy dx \\ &= \int_0^1 \int_0^1 x \left(x - \frac{1}{2}y + \frac{3}{4} \right) dy dx \\ &= \frac{7}{12} \end{aligned}$$

$$\begin{aligned} E(Y) &= \int_0^1 \int_0^1 y \left(x - \frac{1}{2}y + \frac{3}{4} \right) dy dx \\ &= \frac{11}{24} \end{aligned}$$

$$\begin{aligned} E(XY) &= \int_0^1 \int_0^1 xy \left(x - \frac{1}{2}y + \frac{3}{4} \right) dy dx \\ &= \frac{13}{48} \end{aligned}$$

$$\text{Cov}(X, Y) = \frac{13}{48} - \frac{7}{12} \frac{11}{24} = \frac{1}{288}$$

Similar calculations give that $E(X^2) = \frac{5}{12}$ and $E(Y^2) = \frac{7}{24}$, which again gives that $\text{Var}(X) = \frac{5}{12} - (\frac{7}{12})^2 = \frac{11}{144}$ and $\text{Var}(Y) = \frac{7}{24} - (\frac{11}{24})^2 = \frac{47}{576}$ and then we get

$$\rho(X, Y) = \frac{1/288}{\sqrt{(11/144) \cdot (47/576)}} = \underline{\underline{0.04}}$$

i.e. practically no correlation between water consumption (X) and humus content (Y). $\square$

The previous example show that if the joint distribution of two r.v.s is known the exact correlation between the variables can be calculated.

In practice the situation is often that the joint distribution is unknown, but some data (measurements) on joint values of the variables are available. From these measurements it is then possible to calculate the estimated, or *sample* correlation.

Definition - Sample mean and sample variance: Let $X_1, \dots, X_n$ be a random sample from a distribution with finite expectation $\mu = E(X)$ and variance $\sigma^2 = \text{Var}(X)$. The sample mean

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

is an **estimator** of μ and is called the **sample mean**, or **sample average**.

The **sample variance**

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

is an estimator of σ^2 .

Definition – Sample covariance and sample correlation:

Let $(X_1, Y_1), \dots, (X_n, Y_n)$ be a random sample from a joint distribution.

The **sample covariance** is

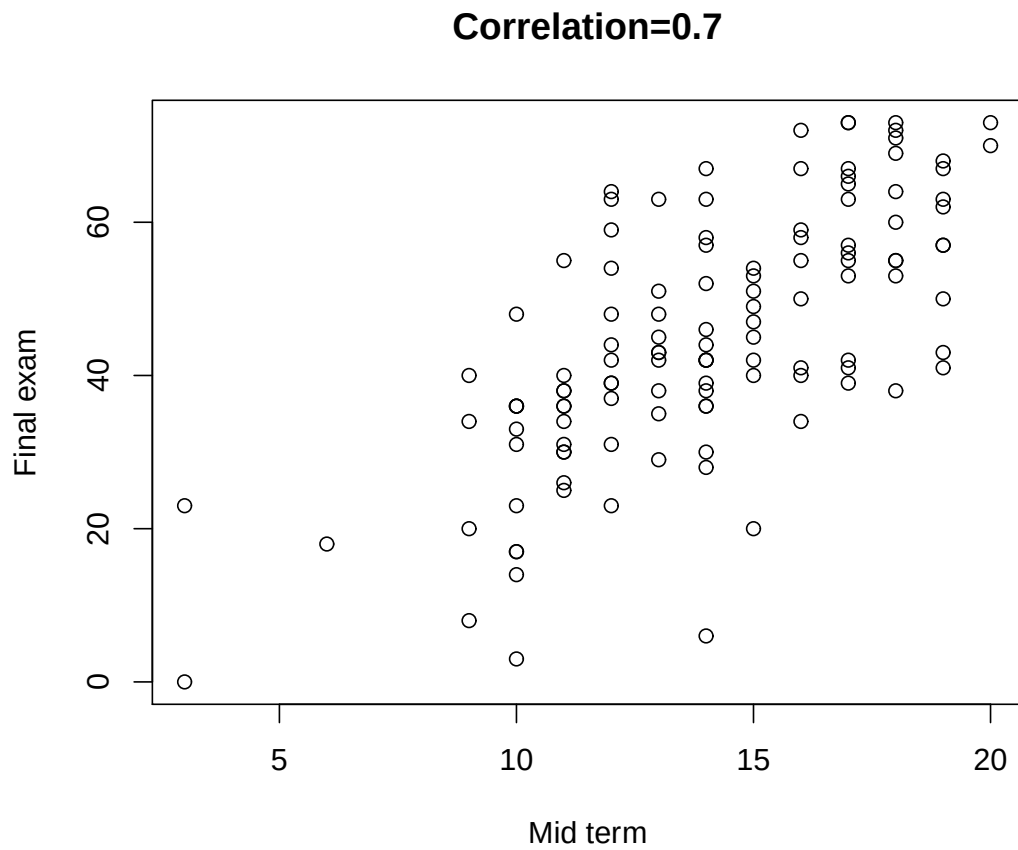
$$S_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}),$$

and the **sample correlation** is

$$R = \frac{S_{XY}}{\sqrt{S_X^2 S_Y^2}}.$$

It satisfies $-1 \leq R \leq 1$.

Example: Results of midterm and final exam.



Expectation and variance of linear combinations

By applying the definitions of expectation and variance to linear combinations of r.v.s we get several rules for linear combinations. Most of these rules are summed up by the following general rule:

Theorem: Let $Y = \sum_{i=1}^n a_i X_i + b$. Then

$$E(Y) = \sum_{i=1}^n a_i E(X_i) + b,$$

$$\text{Var}(Y) = \sum_{i=1}^n a_i^2 \text{Var}(X_i) + 2 \sum_{i=1}^n \sum_{j=1}^{i-1} a_i a_j \text{Cov}(X_i, X_j). \quad \square$$

Note in particular that the rule gives that $E(aX + b) = aE(X) + b$ and that $\text{Var}(aX + b) = a^2 \text{Var}(X)$ (which in particular implies that $\text{Var}(-X) = \text{Var}(X) \dots$).

Also remember that in the case of independence $\text{Cov} = 0$.

Example: We are going to use an instrument for measuring the concentration of a specific substance in a liquid. We know that measurements taken by this instrument give results which have a distribution with expectation μ (true concentration) and variance $\sigma^2 = 0.01$. Instead of doing only one measurement we decide to take 10 measurements and to use the average of these 10 measurements as our result. What is the expectation and variance of this average? Let $\bar{X} = \frac{1}{10}(X_1 + X_2 + \cdots + X_{10}) = \frac{1}{10}X_1 + \cdots + \frac{1}{10}X_{10}$ be that average of the 10 measurement. We assume that the 10 measurements are independent.

$$\begin{aligned}
 E(\bar{X}) &= \frac{1}{10}E(X_1) + \cdots + \frac{1}{10}E(X_{10}) \\
 &= \frac{1}{10}\mu + \cdots + \frac{1}{10}\mu = \mu \\
 \text{Var}(\bar{X}) &= \left(\frac{1}{10}\right)^2\text{Var}(X_1) + \cdots + \left(\frac{1}{10}\right)^2\text{Var}(X_{10}) \\
 &= \left(\frac{1}{10}\right)^2\sigma^2 + \cdots + \left(\frac{1}{10}\right)^2\sigma^2
 \end{aligned}$$

$$= \frac{\sigma^2}{10} = \frac{0.01}{10} = 0.001$$

I.e., the average has the same expectation as single measurements, but lower variance. If we use the rule of thumb saying that around 95% of the results will fall in the interval $\mu \pm 2\sigma$ we get that a single measurement has around 95% probability of falling in the interval $[\mu - 2 \cdot \sqrt{0.01}, \mu + 2 \cdot \sqrt{0.01}] = [\mu - 0.2, \mu + 0.2]$, while the average of 10 measurements has around 95% probability of falling in $[\mu - 2 \cdot \sqrt{0.001}, \mu + 2 \cdot \sqrt{0.001}] = [\mu - 0.06, \mu + 0.06]$.

Discrete probability distributions.

Chapter 2.2 in the textbook.

Recall: $f(x)$ is the pmf (probability mass function) of a discrete r.v. X if:

1. $f(x) \geq 0$
2. $\sum_x f(x) = 1$
3. $f(x) = P(X = x)$

We shall now look at various situations where $f(x)$ is given by a specific parametric function.

Discrete uniform distribution

If X has k possible values $x_1, \dots, x_k$ and every value is attained with the same probability we say that X has a *discrete uniform distribution*.

We then have that

$$f(x) = \frac{1}{k}, x = x_1, \dots, x_k$$

Example: Throw of a dice, $k = 6$, $x_1 = 1, x_2 = 2, \dots, x_6 = 6$. $\square$

Expectation and variance:

$$\mu = E(X) = \sum_x x f(x) = \sum_x x \frac{1}{k} = \frac{1}{k} \sum_x x$$

$$\sigma^2 = \text{Var}(X) = \sum_x (x - \mu)^2 f(x) = \frac{1}{k} \sum_x (x - \mu)^2$$

Example: Throw of a dice.

$$E(X) = \frac{1}{6} \sum_{x=1}^6 x = 3.5$$

$$\text{Var}(X) = \frac{1}{6} \sum_{x=1}^6 (x - 3.5)^2 = 2.92 \quad \square$$

Binomial distribution

In situations characterized by repeated trials where

- The trials are independent.
- We are recording “success” or not “success” in each trial (whether a specific event occurs or not).
- The probability of “success” is the same in each trial, p .
- A specified number of trials, n .

we have that $X =$ “the number of successes in the n trials” has a binomial distribution with parameters n and p . We write $X \sim \text{Bin}(n, p)$, and have that

$$f(x) = \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, \dots, n.$$

Further it can be shown that:

$$E(X) = np \quad \text{and} \quad \text{Var}(X) = np(1 - p)$$

Examples of variables that typically have a binomial distribution:

- The number of components of the same type functioning after time t .
- The number of persons being against EU in a random sample of people.
- The number of patients which gets a certain infection at a hospital unit.
- The number of times we get 6 in n throws of a dice.
- The number of times a the outcome of a simulated random variable exceeds a certain value. □

Multinomial distribution

Generalization of the binomial distribution to situations where we have more than two possible outcomes.

- The trials are independent.
- Each trial has k possible outcomes.
- The probabilities, $p_1, \dots, p_k$ for each of the k outcomes are the same in all trials.
- A specified number of trials, n .

Then the joint distribution of $X_1, \dots, X_k$, where X_j represents the number of times outcome j occurs, is a *multinomial distribution* with parameters n and $p_1, \dots, p_k$. We then have (can be shown):

$$f(x_1, \dots, x_k) = \frac{n!}{x_1! \cdots x_k!} p_1^{x_1} \cdots p_k^{x_k}$$

where $\sum_{i=1}^k x_i = n$ and $\sum_{i=1}^k p_i = 1$. It can also be shown that

$$\begin{aligned} E(X_i) &= np_i, \quad \text{Var}(X_i) = np_i(1 - p_i), \\ \text{Cov}(X_i, X_j) &= -np_i p_j \text{ for } i \neq j. \end{aligned}$$

Negative binomial distribution

If we have

- Independent trials.
- “Success” or not “success” in each trial.
- The probability of “success” is the same in each trial, p .
- Repeat the trials until a specified number of successes, r , are observed.

and $Y =$ “the number of trials until success number r ”, then Y is having a *negative binomial distribution* with parameters r and p .

Note: In a binomial experiment we look at the number of successes in a given number of trials (the number of successes is random), while in a negative binomial experiment we look at the number of trials needed to get a specific number of successes (the number of trials is random).

It can be shown by a similar reasoning as for the binomial distribution that:

$$f(y) = \binom{y-1}{r-1} p^r (1-p)^{y-r}, \quad y = r, r+1, \dots$$

and that

$$E(Y) = \frac{r}{p}, \quad \text{Var}(Y) = r \frac{1-p}{p^2}$$

Examples of variables that typically have a negative binomial distribution:

- The number of components which needs to be fabricated until we have r components without failures.
- The number of houses a vacuum cleaner seller needs to visit until he has sold r vacuum cleaners.
- The number of persons you have to ask before you get r signatures to a signature campaign.
- The number of times you have to throw a dice until you get 6 for the r th time. $\square$

Example: At a china (porcelain) factory all products goes through a quality control. Only products classified as A-products are sent to ordinary sale. For a certain kind of plates the factory has experienced that 91% of the produced plates can be classified as A-products. The plates are packed in boxes containing 6 plates. What is the probability that we have to check 8 or more plates before we have enough plates classified as A-products to fill a box? What is the expected number of plates we need to check to fill a box?

$$\begin{aligned}
 P(Y \geq 8) &= 1 - P(Y \leq 7) = 1 - [P(Y = 6) + P(Y = 7)] \\
 &= 1 - \binom{6-1}{6-1} 0.91^6 (1 - 0.91)^{6-6} \\
 &\quad - \binom{7-1}{6-1} 0.91^6 (1 - 0.91)^{7-6} \\
 &= 1 - 0.5678 - 0.3066 = \underline{\underline{0.126}}
 \end{aligned}$$

$$E(Y) = \frac{r}{p} = \frac{6}{0.91} = \underline{\underline{6.6}} \quad \square$$

See R-code: `neg_bin_demo.R`

Alternative formulation

Sometimes the random variable is defined as (see the book) $X =$ “the number of failures until success number r ”, then $X = Y - r$ and

$$f(x) = \binom{x+r-1}{r-1} p^r (1-p)^x, \quad x = 0, 1, 2, \dots$$

and

$$E(X) = r \frac{1-p}{p}, \quad \text{Var}(X) = r \frac{1-p}{p^2}$$

Geometric distribution

A special case of the negative binomial distribution where we look at the number of trials until the *first* success, i.e. $r = 1$.

E.g. the number of times an oil drilling company needs to drill at a certain location until the first time they find oil.

Poisson distribution

Typical for a *Poisson experiment* is that we look at the number of times an event occur inside a time interval or specific area.

Examples of variables that typically are Poisson distributed:

- The number of phone calls to an office during one hour.
 - The number of visitors to a web page per hour.
 - The number of traffic accidents on a specific road during one year.
 - The number of times a fire brigade is called out per week.
 - The number of bugs page per 1000 lines of program code.
 - The number of trees per acres in a forest.
-

The occurrence of events follow a *Poisson process* if:

1. The numbers of events in non-overlapping intervals are independent.
2. For any small interval $[s, s + \Delta s]$ we have that $P(\text{one event in } [s, s + \Delta s]) \approx \lambda \Delta s$.
3. $P(\text{two or more events in } [s, s + \Delta s]) \approx 0$.

If this is fulfilled $X =$ "number of events in $[0, t]$ " is Poisson distributed with parameter $\mu = \lambda t$. It can then be shown that:

$$f(x) = \frac{(\lambda t)^x}{x!} e^{-\lambda t}, \quad x = 0, 1, 2, \dots$$

and that $\mu = E(X) = \lambda t$ and $\text{Var}(X) = \lambda t$.

The parameter λ represents the expected number of events per unit and is also called the *intensity* of the process.

We shall learn more about Poisson-processes later in the course.

Continuous probability distributions.

Chapter 2.3 in the textbook.

Recall: $f(x)$ is the pdf (probability density function) of a continuous r.v. X if:

1. $f(x) \geq 0$ for all $x \in \mathcal{R}$
2. $\int_{-\infty}^{\infty} f(x)dx = 1$
3. $P(a < X < b) = \int_a^b f(x)dx$

We shall now look at some of the most commonly used continuous probability distributions. (For each such distribution $f(x)$ is given by a specific parametric function.)

Continuous uniform distribution

If X can only attain values in an interval $[a, b]$, and it is the same probability that X attains a value in any interval of the same length inside $[a, b]$, then X is having a uniform distribution (e.g. the location of a pipe burst). Then

$$f(x) = \frac{1}{b-a}, \quad a \leq x \leq b.$$

and it can be shown that $E(X) = \frac{(a+b)}{2}$ and $\text{Var}(X) = \frac{(b-a)^2}{12}$.

The most common use of the continuous uniform distribution is for simulations. Basic random number generators are generating uniformly distributed numbers between 0 and 1. These can further be used to generate numbers from other probability distributions by various techniques that we shall learn about in this course.

Normal distribution

The normal distribution is the most important and most used of all probability distributions. A large number of different phenomena in nature, technology, economy, etc can be modeled by a normal distribution. The normal distribution is also very important in statistical theory. For instance can many distributions under certain conditions be approximated by the normal distribution.

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left(-\frac{1}{2} \frac{(x - \mu)^2}{\sigma^2} \right) , \quad -\infty < x < \infty,$$

$$E(X) = \mu \text{ and } \text{Var}(X) = \sigma^2.$$

We write $X \sim N(\mu, \sigma^2)$.

Since $f(x)$ is complicated to integrate, we calculate probabilities in the normal distribution by doing the transformation

$$Z = \frac{X - E(X)}{\sqrt{\text{Var}(X)}} = \frac{X - \mu}{\sigma}$$

where Z is having a *standard normal distribution* (check yourself that $E(Z) = 0$ and $\text{Var}(Z) = 1$) and then using a table of the standard normal distribution.

E.g.

$$\begin{aligned} P(X < a) &= P\left(\frac{X - \mu}{\sigma} < \frac{a - \mu}{\sigma}\right) \\ &= P\left(Z < \frac{a - \mu}{\sigma}\right) \end{aligned}$$

where $P(Z < \frac{a-\mu}{\sigma})$ is found from a table. (We do not go into further details on this here, this is assumed known from earlier courses.)

Gamma distribution

The gamma distribution is a flexible distribution defined on $x \geq 0$. The gamma distribution is among other things used to describe lifetimes both for technical systems and in biology.

Several commonly used distributions are special cases of the gamma distribution.

$$f(x) = \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}, \quad x \geq 0$$

$$E(X) = \alpha\beta \text{ and } \text{Var}(X) = \alpha\beta^2.$$

$\Gamma(\alpha)$ is the gamma function. See the textbook. Note in particular that $\Gamma(n) = (n-1)!$ if n is integer.

There are different ways of specifying the parameters of the gamma distribution. The version above is used in R with α called shape parameter and β called scale parameter. However, the book uses a different version where $\alpha = r$ and $\beta = 1/\lambda$.

Example: The lifetime in months of a certain species of insects is well described by a gamma distribution with parameters $\alpha = 2$ and $\beta = 0.5$. What is the probability that an insect of this kind lives for more than 2 months?

$$\begin{aligned} P(X > 2) &= \int_2^{\infty} \frac{1}{0.5^2 \Gamma(2)} x^{2-1} e^{-x/0.5} dx \\ &= \int_2^{\infty} 4x e^{-2x} dx = \underline{\underline{0.092}} \quad \square \end{aligned}$$

See R-code: `gamma_demo.R`

The exponential distribution

An important special case of the gamma distribution is the exponential distribution which we get by setting $\alpha = 1$ in the gamma distribution:

$$f(x) = \frac{1}{\beta} e^{-x/\beta}, \quad x \geq 0,$$

$$E(X) = \beta \text{ and } \text{Var}(X) = \beta^2.$$

Note: Often the parameter $\lambda = 1/\beta$ is used. This is the parameterization used in the book

The exponential distribution is among other things used to model lifetimes (e.g. for electronic components), times until and between events, etc.

See R-code: `exponential_distribution_demo.R`

χ^2 -distribution and t -distribution

Mostly used in statistical theory for confidence intervals and hypothesis testing. See the book.

Beta distribution

The beta distribution is a flexible distribution defined on $[0, 1]$ which among other things is popular for simulations.

$$f(x) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1 - x)^{\beta-1}, \quad 0 \leq x \leq 1$$

$$E(X) = \frac{\alpha}{\alpha + \beta} \text{ and } \text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}.$$

The term $\frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)}$ is the reciprocal of the beta function, see the textbook.

The uniform distribution is the special case $\alpha = \beta = 1$.

Triangle distribution

The triangle distribution is popular for simulations, e.g. to approximate situations with limited knowledge.

$$f(x) = \begin{cases} \frac{2(x-a)}{(b-a)(c-a)} & , a \leq x \leq c \\ \frac{2(b-x)}{(b-a)(b-c)} & , c < x \leq b \\ 0 & , \text{otherwise} \end{cases}$$

$$E(X) = (a + b + c)/3 \quad \text{and}$$

$$\text{Var}(X) = (a^2 + b^2 + c^2 - ab - ac - bc)/18.$$

Sometimes a notation like $a = x_{\min}$, $b = x_{\max}$, $c = x_{\text{most probable}}$ is used.

Lognormal distribution

X is having a lognormal distributed if $\ln(X)$ is having a normal distribution. If $E(\ln(X)) = \mu$ and $\text{Var}(\ln(X)) = \sigma^2$ then

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma x} \exp\left(-\frac{1}{2} \frac{(\ln(x) - \mu)^2}{\sigma^2}\right), \quad x > 0,$$

$E(X) = e^{\mu + \sigma^2/2}$ and $\text{Var}(X) = e^{2(\mu + \sigma^2)} - e^{2\mu + \sigma^2}$. The lognormal distribution is among other things used as a model for concentrations, lifetimes, fatigue, etc

Example: The concentration of a pollutant in the water from a chemical plant is having a lognormal distributed with parameters $\mu = 3.2$ and $\sigma = 1$ (measured in i p.p.m.). What is the probability that the concentration exceeds 8?

$$\begin{aligned} P(X > 8) &= 1 - P(X \leq 8) \\ &= 1 - P(\ln(X) \leq \ln(8)) \\ &= 1 - P\left(\frac{\ln(X) - 3.2}{1} \leq \frac{\ln(8) - 3.2}{1}\right) \\ &= 1 - P(Z \leq -1.12) \\ &= 1 - 0.1314 = \underline{\underline{0.8686}} \quad \square \end{aligned}$$

Multivariate normal distribution

Chapter 2.4 in the textbook.

Among all joint distributions the multivariate normal is by far the most used. It is used a lot in statistical theory, and to model situations when several normally distributed variables are recorded on the same “unit”. E.g. several measurements on the same person or several physical quantities measured on each product.

An important special case is the bivariate normal, for which the density can be written:

$$f(x_1, x_2) = \frac{1}{2\pi|\boldsymbol{\Sigma}|^{1/2}} \exp\{-0.5(\mathbf{x}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\}$$

Here:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad \boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \quad \boldsymbol{\Sigma} = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix}$$

See R-code: `bivariate_normal.R`

von Mises–Fisher distribution

If observations represent directions rather than unrestricted vectors, they lie on the unit sphere

$$S^{p-1} = \{\mathbf{x} \in \mathbb{R}^p : \|\mathbf{x}\| = 1\}.$$

A standard probability model for directional data is the *von Mises–Fisher distribution*. We say $\mathbf{X} \sim vMF(\boldsymbol{\mu}, \kappa)$, where $\boldsymbol{\mu} \in S^{p-1}$ and $\kappa \geq 0$ if

$$f(\mathbf{x}; \boldsymbol{\mu}, \kappa) = c_p(\kappa) \exp \{ \kappa \boldsymbol{\mu}^T \mathbf{x} \}, \quad \mathbf{x} \in S^{p-1}.$$

Here $\boldsymbol{\mu}$ is the **mean direction** and κ is the **concentration parameter**. Larger values of κ give stronger concentration around $\boldsymbol{\mu}$, while $\kappa = 0$ gives the uniform distribution on the sphere.

The normalizing constant is

$$c_p(\kappa) = \frac{\kappa^{p/2-1}}{(2\pi)^{p/2} I_{p/2-1}(\kappa)},$$

where I_ν is the modified Bessel function of the

first kind, defined by

$$I_\nu(\kappa) = \sum_{m=0}^{\infty} \frac{1}{m! \Gamma(m + \nu + 1)} \left(\frac{\kappa}{2}\right)^{2m+\nu}.$$

For $p = 2$ the von Mises–Fisher distribution reduces to the *von Mises distribution* on the unit circle, while for $p = 3$ it is often simply called the *Fisher distribution*.

Special case $p = 2$: the von Mises distribution on the unit circle S^1 .

$$f(\theta; \mu, \kappa) = \frac{1}{2\pi I_0(\kappa)} \exp\{\kappa \cos(\theta - \mu)\}, \quad 0 \leq \theta < 2\pi.$$

Special case $p = 3$: the Fisher distribution on the unit sphere S^2 .

$$f(\mathbf{x}; \boldsymbol{\mu}, \kappa) = \frac{\kappa}{4\pi \sinh(\kappa)} \exp\{\kappa \boldsymbol{\mu}^T \mathbf{x}\}, \quad \mathbf{x} \in S^2.$$

See R-code: `MisesFisher.R`